Calvin Gacheru Mwangi

Data Analyst · Data Scientist · AI Engineer

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Python LangChain LangGraph scikit-learn FastAPI Docker PostgreSQL SQL RAG MLOps

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Summary

Fourth-year Data Science & Analytics student who ships production software, not just notebooks. I've built a behavioral RAG chatbot, an agentic AI productivity assistant, and predictive machine learning models — plus full-stack platforms taken from domain registration to live deployment. I own the complete lifecycle: cloud deployments, ETL pipelines, CI/CD, and security-hardened infrastructure. I don't just train models — I ship them into production.

Core Skills

Languages & Frameworks
Python · SQL · HTML / CSS / JS · R · Bash · Linux (Fedora)

AI / Machine Learning
scikit-learn · LangChain · LangGraph · RAG Architectures · OpenAI API · Vector Databases · Agentic Orchestration · NLP · Time Series Analysis · Predictive Analytics · NumPy · Pandas · statsmodels

Engineering & Infrastructure
Docker · Kubernetes · FastAPI · Flask · Django · PostgreSQL · Redis · Git / GitHub · CI/CD (GitHub Actions) · Render

Other
ETL Pipelines · Exploratory Data Analysis · Power BI · Streamlit · Cloud Security (TLS/SSL, IAM/RBAC)

Experience

BikBat Insurance Agency — Freelance Web Developer *(Live Client · 2026)*

- Delivered the full business website end-to-end for a live client — domain registration, DNS configuration, and cloud hosting — as a standalone engagement from brief to production.
- Built a brochure site with service listings and trust-building content designed to convert visitors into leads. bikbatinsurance.com

ProdigyInfoTech — Data Science Virtual Internship *(2025)*

- Completed four end-to-end data science projects spanning exploratory analysis, predictive modeling, customer churn analytics, and sentiment analysis on real-world datasets.
- Produced documented, reproducible analyses for every task: EDA · Predictive Analytics · Customer Churn · Sentiment Analysis

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Featured Projects

MySaidi — AI Productivity Assistant

Full-stack AI assistant that manages tasks, calendars, and habits through natural conversation. A LangGraph agent running on Groq parses user intent and directly edits the user's schedule. Shipped as a secure, installable Progressive Web App with JWT authentication.

Stack: Python · FastAPI · LangGraph · Groq LLM · PostgreSQL · JWT Auth · PWA
github.com/Calvin-Gacheru/MySaidi

Uber Price Prediction & MLOps Pipeline

Production-grade MLOps pipeline predicting Uber ride prices with a tuned Random Forest model. Served through a FastAPI + Pydantic backend behind a Streamlit UI, containerized and shipped with automated CI/CD — a repeatable path from raw data to deployment.

Stack: Python · scikit-learn · FastAPI · Docker · Streamlit · GitHub Actions
github.com/Calvin-Gacheru/Uber-Price-Prediction

AI-Powered Knowledge Retrieval System

Full-stack RAG application that streamlines technical support by centralizing documentation. Ingests and embeds PDFs into pgvector, then uses Groq via LangChain to generate context-aware answers — Dockerized end-to-end with a built-in QA feedback loop.
Stack: Python · FastAPI · LangChain · Groq LLM · pgvector · Docker
github.com/Calvin-Gacheru/AI-KNOWLEDGE-RETRIEVAL-SYSTEM

Diabetes Risk Screening Tool

Recall-optimized screening pipeline for early diabetes detection, tackling severe class imbalance with SMOTE and a class-weighted XGBoost classifier. **Achieves 88% recall** and ships as a Streamlit app with live feature engineering on raw clinical inputs.
Stack: Python · XGBoost · SMOTE · scikit-learn · Streamlit
github.com/Calvin-Gacheru/Diabetes-Prediction

Twitter Entity Sentiment Analysis

NLP pipeline tracking public sentiment toward brands from X (Twitter) posts in real time. Tuned the classifier via GridSearchCV to a **93% F1-score** and shipped it as a live Streamlit app for on-demand brand monitoring.
Stack: Python · NLP · scikit-learn · GridSearchCV · Streamlit
github.com/Calvin-Gacheru/Twitter-Tweets-Sentimental-Analysis

Secure Cloud Storage Architecture

Hardened MinIO object-storage environment in Docker demonstrating enterprise security controls: TLS-encrypted transport, RBAC via custom IAM policies, and server-side encryption with a KMS key. Validated against a synthetic dataset of **100,000 corporate records**.
Stack: Python · MinIO · Docker · TLS/SSL · IAM/RBAC
github.com/Calvin-Gacheru/Secure-Cloud-Storage

Additional Projects

Project	What it does	Stack	Link
Behavioural Explainer	Local-first RAG system analyzing human-interaction scenarios	Ollama, Llama 3.1, Qdrant, RAG	Repo

	through a psychological lens		
Maternal Risk Detection System	Flags high-risk pregnancies from patient vitals; SHAP explains predictions; automated SMS reminders	Python, SHAP, APScheduler, SQLite, Streamlit	Repo
Stroke Prediction Using Bayesian Networks	Discrete Bayesian Network estimating stroke risk via probabilistic reasoning and exact inference	Python, pgmpy, pandas	Repo
Customer Subscription Prediction	Predicts bank term-deposit subscriptions; tuned LightGBM reaches 88% recall	Python, LightGBM, Power BI, GridSearchCV	Repo
Lung Cancer Survival Prediction	Predicts survival outcomes from clinical records with feature engineering and survival analysis	Python, scikit-learn, lifelines, Plotly	Repo
ETL Data Engineering	Custom ETL pipelines with validation, logging, and scheduling for structured data	Python, SQL, Pandas	Repo
Exploratory Data Analysis	Cross-dataset analysis surfacing patterns and business-ready insights	Python, Pandas, Power BI, Matplotlib	Repo
Global Air Quality Analysis	Tracks pollutant trends via OpenAQ API; forecasts AQI with scikit-learn	Python, Pandas, Seaborn, scikit-learn	Repo
Arcade Kenya	Live, multi-game entertainment platform with rich animation and brand identity	JavaScript, HTML/CSS	Live · Repo
Valen Time Love	Bespoke, personalized web application from a custom client brief	JavaScript, HTML/CSS	Live
WebCraft — Website Brochure	Responsive brochure-site template for fast customization and deployment	HTML/CSS/JS	Live

Certifications

- **Building AI Agents using LangGraph** — Simplilearn — LangGraph workflows, stateful orchestration, practical agent implementation
- **Data Science Virtual Internship** — ProdigyInfoTech (2025) — EDA, machine learning, churn analytics, sentiment analysis

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Education

BSc. Data Science & Analytics
United States International University — Africa (USIU-Africa)
Year 4 of 4 · In Progress · Expected Graduation 2027

Soft Skills

Analytical & Critical Thinking · Business-Minded Problem Solving · Cross-Functional Communication · Autonomous & Fast Learning · Ownership & End-to-End Delivery · Stakeholder Management

Languages

English (Native) · Swahili (Native)

Interests

Music · Linux / Systems Tinkering · Kaggle & Data Competitions · Reading (Tech & Non-Fiction) · Chess & Strategy Games